

Adaptive Working Condition Recognition With Clustering-Based Contrastive Learning for Unsupervised Anomaly Detection

Qifa Xu , Tianming Xie , Cuixia Jiang , Qiliang Cheng , and Xiangxiang Wang 

Abstract—In real industrial processes, machines usually run under variable working conditions, which impose challenges for anomaly detection. To complete anomaly detection for machines under variable working conditions, we develop a reconstruction-based autoencoder called clustering-based contrastive learning autoencoder (CBCL-AE). It integrates clustering-based contrastive learning (CBCL) to perform clustering in the feature space and enhance the differentiation of features from different working conditions, thereby achieving adaptive working condition recognition. Considering the crucial role of the clustering of CBCL, we theoretically and experimentally demonstrate its convergence property during the training process, which directly determines the effectiveness of CBCL-AE. CBCL-AE's superiority has been validated on three public datasets and two private datasets collected from an actual industrial process. These validations highlight its superiority over five state-of-the-art models in unsupervised anomaly detection.

Index Terms—Adaptive working condition recognition, anomaly detection, clustering-based contrastive learning (CBCL), multivariate time series (MTS), reconstruction-based autoencoder, variable working conditions.

I. INTRODUCTION

IN THE era of Industry 4.0, the significance of anomaly detection is growing exponentially across various industries, encompassing wind power, transportation, oil, natural gas, aerospace, vehicles, etc. [1], [2]. As the earliest stage of Prognostics Health Management, anomaly detection is capable

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Qifa Xu and Cuixia Jiang are with the School of Management, Hefei University of Technology, Hefei 230009, China (e-mail: xuqifa@hfut.edu.cn; jiangcuixia@hfut.edu.cn).

Tianming Xie is with the School of Management, Hefei University of Technology, Hefei 230009, China, and also with the Faculty of Engineering and Environment, Northumbria University, NE1 8ST Newcastle upon Tyne, U.K. (e-mail: xietianming@mail.hfut.edu.cn).

Qiliang Cheng and Xiangxiang Wang are with RONS Science and Technology Incorporated Company, Hefei 230000, China (e-mail: qil-
iang.cheng@ronds.com.cn; xiangxiang.wang@ronds.com.cn).

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of promptly capturing abnormal behaviors during machine operations. This lays a crucial foundation for subsequent machine fault diagnosis and repair processes. To date, there have been numerous studies on anomaly detection using various types of data, including images [3], videos [4], multivariate time series (MTS) [5], and so on. In general, MTS can retain more information in the temporal dimension than in images and videos with the same storage capacity. As a result, MTS anomaly detection has been increasingly applied in various industrial production environments as a cost-effective solution to reduce storage costs.

In real industrial processes, the existing anomaly detection methods can be primarily grouped into three types: model-based, signal-based, and knowledge-based methods. First, model-based methods [6], [7], [8] leverage prior information from industrial processes or practical system models, enabling accurate anomaly detection even with limited data. However, it is difficult to acquire accurate, comprehensive, and reliable models in complicated industrial production environments. Moreover, the system's outputs are instability and may be influenced by various factors, such as environmental noise, machine lubrication, and temperature, which reduce the efficiency of model-based methods. Second, signal-based methods [9] extract features from measured signals, then analyze symptoms and make diagnostic decisions based on prior knowledge of the system. Owing to the rich information contained in time-domain and frequency-domain signals, the performance of fault diagnosis in motors [10] and rotary machines [11] can be improved. For this reason, signal-based anomaly detection requires high storage capacity to preserve more detailed information. Third, the characteristic of knowledge-based methods is to extract underlying knowledge from a large volume of historical data using artificial intelligence techniques. The knowledge-based method [12], [13], [14] is also referred to as a data-driven method for its reliance on historical data for decision-making. With the rapid progress in big data and computing capability, knowledge-based methods that rely on artificial intelligence models have gained significant attention as a popular research area in recent years.

In industrial production environments, a machine anomaly is a typical low-probability event and presents some characteristics, such as unpredictability, unknown patterns, and diversity. Therefore, unsupervised data-driven methods are more suitable for anomaly detection. In these methods, autoencoders [15] are usually employed to construct reconstruction-based models,

which aim at differentiating between abnormal and normal behaviors by learning patterns or features from normal behaviors. Wulsin et al. [16] applied deep belief networks to construct an encoding–decoding model, which utilizes reconstruction error to detect anomalies in clinical electroencephalography data. However, without considering temporal dependencies, such encoding–decoding models [17] and [18] are limited significantly in terms of accuracy in unsupervised anomaly detection. To take temporal dependencies into account, many studies use recurrent neural networks (RNNs) [5], [19], [20], including traditional RNNs, long short-term memory (LSTM), and gate recurrent unit (GRU), to construct autoencoder. These studies have demonstrated that such models excel in reconstructing MTS data. Indeed, in real-world data, it is common to encounter various noises that can potentially decrease the robustness of the model during the training process, even in normal data. To address this issue, the authors in [5], [20], and [21] incorporated variational inference [22] into the autoencoder. By introducing variational inference, the autoencoder can represent features in the form of distributions, reducing the impact of noises at the tails of the distribution on the model significantly. In recent years, the transformer has also been gradually introduced into MTS anomaly detection since its powerful text processing capability. Based on the characteristics of the transformer, Xu et al. [23] constructed two types of associations: the association between the current point and local context, as well as the association between the current point and the global sequence. As a result, the distinguishability between normal data and anomalies has been successfully increased. The self-attention mechanism in the transformer calculates temporal dependencies with parallel matrix multiplication operations, which enables the transformer to run much faster than RNNs with the same number of parameters. However, this computational advantage in terms of time is accompanied by an increased storage burden. In addition to the methods mentioned previously, there are also autoencoders built by integrating generative adversarial networks [24] or combined with graph neural networks [25], [26]. The former aims at enhancing the model’s generation or data reconstruction capabilities, while the latter focuses on establishing better correlations among variables in high-dimensional data. Current research on anomaly detection is primarily focused on steady-state or single-machine scenarios. However, due to the complex industrial processes, some machines often work under variable conditions. In fact, for machines under variable working conditions, having a large amount of data with such different working conditions in the training set might enhance the model’s ability to reconstruct genuine anomalies. Variable working conditions lead to the characteristic of concept drift in multisensor time series data, where the distribution, such as mean and/or variance, of the MTS changes abruptly over time. This makes it more difficult to identify true anomalies. Research on anomaly detection specifically for machines under variable working conditions is scarce, which remains a significant challenge within the industrial sector.

To this end, we develop a reconstruction-based autoencoder called clustering-based contrastive learning autoencoder (CBCL-AE). It integrates clustering-based contrastive learning (CBCL) to perform clustering in the feature space and enhance

the differentiation of features from different working conditions, thereby achieving adaptive working condition recognition. The contributions of this study can be summarized as follows.

- 1) We integrate CBCL into a specific autoencoder built with GRU and realize end-to-end training. It can adaptively group similar working condition data into the same category with the clustering method. By leveraging contrastive learning, the CBCL-AE model can effectively improve the proximity of features from identical working conditions. It can also enhance the differentiation of features from different working conditions by designing a specific constraint to ensure a sufficient distance among these features.
- 2) We conclude that the clustering process in the feature space can converge by theoretically proving the model’s convergence during the training process, as the clustering process relies on model optimization. The proof of convergence of this clustering process provides theoretical support for the effectiveness of the integrated CBCL module in CBCL-AE. By experimentally visualizing the trace of cluster center iteration during the training process, we further confirm the convergence of the clustering process.
- 3) We conduct experiments on three public datasets and two private datasets. The comparison results on three public datasets demonstrate that the CBCL-AE model achieves acceptable results in general MTS anomaly when compared to five state-of-the-art models (SOTAs). The main results on our private datasets show that our model is superior to the five SOTAs in monitoring machines under variable working conditions. In addition, the ablation study proves the efficacy of integrating CBCL into our model.

The rest of this article is organized as follows. Section II introduces relevant theories and definitions. Section III details the proposed CBCL-AE model. Section IV considers real-world applications on three public datasets and two private datasets. Finally, Section V concludes this article.

II. PRELIMINARIES

A. Reconstruction-Based Anomaly Detection

Due to the unpredictability, unknown patterns, and diversity of anomalies in MTS, supervised methods are in general not suitable for MTS anomaly detection. To this end, we develop a reconstruction-based model using an autoencoder to realize unsupervised anomaly detection. Let us consider an MTS $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_t, \dots, \mathbf{x}_T]$ with a length of T , where each element $\mathbf{x}_t \in \mathbb{R}^m$ is an m -dimensional vector. Our goal is to build a reconstruction-based model, denoted as $f(\cdot)$, which takes $\mathbf{X}$ as inputs and produces $\hat{\mathbf{X}}$ as outputs. The model aims to maximize the similarity between $\mathbf{X}$ and $\hat{\mathbf{X}}$. Consequently, the residuals $\mathbf{e} = \mathbf{X} - \hat{\mathbf{X}}$ can be used to identify anomalies.

B. Contrastive Learning

The core idea of contrastive learning is to bring observations from the same category closer together in the feature space, while simultaneously increasing the separation between observations

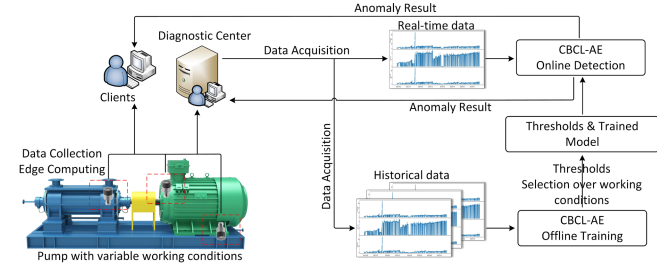


Fig. 1. Overall architecture of unsupervised anomaly detection based on CBCL-AE.

from different categories. This can be achieved by minimizing the distance between observations within the same category and maximizing the distance between observations from different categories. This nature of contrastive learning can help improve the quality of feature representations.

Following Hadsell et al.'s [27] work, we introduce a traditional contrastive learning method with the objective function

$$L_C(\theta) = \frac{1}{N} \sum_{n=1}^N \left(yD(f_\theta(X_1^{(n)}), f_\theta(X_2^{(n)}))^2 + (1-y) \max(\lambda - D(f_\theta(X_1^{(n)}), f_\theta(X_2^{(n)})), 0)^2 \right) \quad (1)$$

where N is the batch size, y indicates whether X_1 and X_2 belong to the same category, $D(\cdot)$ represents the distance calculation function, θ is the parameter set of the network $f_\theta(\cdot)$, and λ is a tunable parameter to control the minimum distance that should be maintained between different categories. In addition, we will present the integration of CBCL into the model in Section III-B.

III. CLUSTERING-BASED CONTRASTIVE LEARNING AUTOENCODER

A. Overall Architecture

1) *Scheme*: Fig. 1 presents the overall architecture of unsupervised anomaly detection based on CBCL-AE. First, data from sensors installed on different components of a pump under variable working conditions are collected using edge computing techniques. Second, the data are transmitted to both the clients and servers of the diagnostic center, allowing access to historical data for offline training of CBCL-AE. Third, thresholds for different working conditions are selected. Finally, based on these chosen thresholds and the trained model, anomalies in real-time data can be effectively detected and the results are promptly communicated back to the clients.

2) *Architecture*: Fig. 2 shows the architecture of the proposed CBCL-AE model. It integrates the contrastive learning method and requires that model's inputs are a pair of observations. This pair of observations is randomly sampled from the training set, which means they could either come from the same working condition or different working conditions. This pair of inputs, $\mathbf{X}$ and $\mathbf{X}'$, will be fed to an autoencoder built with RNN. Similar

to LSTM, GRU is capable of alleviating the issue of gradient vanishing and has a simpler architecture. Therefore, we employ GRU to construct the autoencoder. In Fig. 2, EU and DU denote the encoder unit and the decoder unit that consist of GRU, which are defined as

$$\begin{aligned} \mathbf{r}_t &= \sigma(\mathbf{W}_{rx}\mathbf{x}_t + \mathbf{W}_{rh}\mathbf{h}_{t-1} + \mathbf{b}_r) \\ \mathbf{z}_t &= \sigma(\mathbf{W}_{zx}\mathbf{x}_t + \mathbf{W}_{zh}\mathbf{h}_{t-1} + \mathbf{b}_z) \\ \tilde{\mathbf{h}}_t &= \phi(\mathbf{W}_x\mathbf{x}_t + \mathbf{W}_h(\mathbf{r}_t \odot \mathbf{h}_t) + \mathbf{b}_h) \\ \mathbf{h}_t &= (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \end{aligned} \quad (2)$$

where $\mathbf{r}_t$ and $\mathbf{z}_t$ are the reset gate vector and update gate vector in GRU, respectively; $\tilde{\mathbf{h}}_t$ represents the candidate activation vector; $\mathbf{h}_t$ refers to the output vector of a GRU; $\mathbf{x}_t$ is the input at time step t , while $\mathbf{W}$ represents the weight vectors in GRU. In addition, $\sigma(\cdot)$ denotes the sigmoid activation function and $\phi(\cdot)$ is the tanh activation function.

To construct the GRU autoencoder (GRU-AE), we adopt four layers of GRU, with two layers employed for the encoder and two layers for the decoder. The reconstruction process of the GRU-AE can be presented as follows:

$$\mathbf{h}_t^{(l)} = \begin{cases} EU_l(\mathbf{h}_t^{(l-1)}, \mathbf{h}_{t-1}^{(l)}), & l \leq \frac{L}{2} \\ DU_l(\mathbf{h}_t^{(l-1)}, \mathbf{h}_{t-1}^{(l)}), & \frac{L}{2} < l \leq L \end{cases} \quad (3)$$

where $\mathbf{h}_t^{(l)}$ represents the hidden state of the l th layer; $EU_l(\cdot)$ denotes the encoder layer corresponding to the l th layer in GRU-AE, and $DU_l(\cdot)$ follows the same representation; Specific expression of $EU_l(\cdot)$ and $DU_l(\cdot)$ can be referred to the calculation of $\mathbf{h}_t$ in (2); L is the total number of layers in the GRU-AE ($L = 4$ for the proposed model). In particular, when $L = 1$, the input $\mathbf{h}_t^{(l-1)}$ in $EU_l(\cdot)$ is actually the model's input $\mathbf{x}_t$; when $L = 4$, the output hidden state $\mathbf{h}_t$ is the reconstruction $\hat{\mathbf{x}}_t$. However, GRU-AE only possesses the ability to reconstruct MTS and is unable to achieve adaptive condition recognition. Therefore, we further introduce CBCL in GRU-AE.

B. Clustering-Based Contrastive Learning

Denote the hidden states of the pair of inputs $\mathbf{X}$ and $\mathbf{X}'$, obtained by the encoder at time t , as $\mathbf{h}_t^{(2)}, \mathbf{h}_t'^{(2)}$, which can also be considered as the extracted features in the feature space. According to (1), the objective function of the contrastive learning can be written as

$$L_C(\theta) = \frac{1}{N} \sum_{n=1}^N \left(yD(\mathbf{h}_{tn}^{(2)}, \mathbf{h}_{tn}'^{(2)})^2 + (1-y) \max\left(\left(\lambda - D(\mathbf{h}_{tn}^{(2)}, \mathbf{h}_{tn}'^{(2)})\right), 0\right)^2 \right). \quad (4)$$

Since it is unknown whether $\mathbf{X}$ and $\mathbf{X}'$ belong to the same working condition, i.e., y is unknown, we obtain y through a k-means clustering algorithm of $\mathbf{h}_t$ in a batch during the training process.

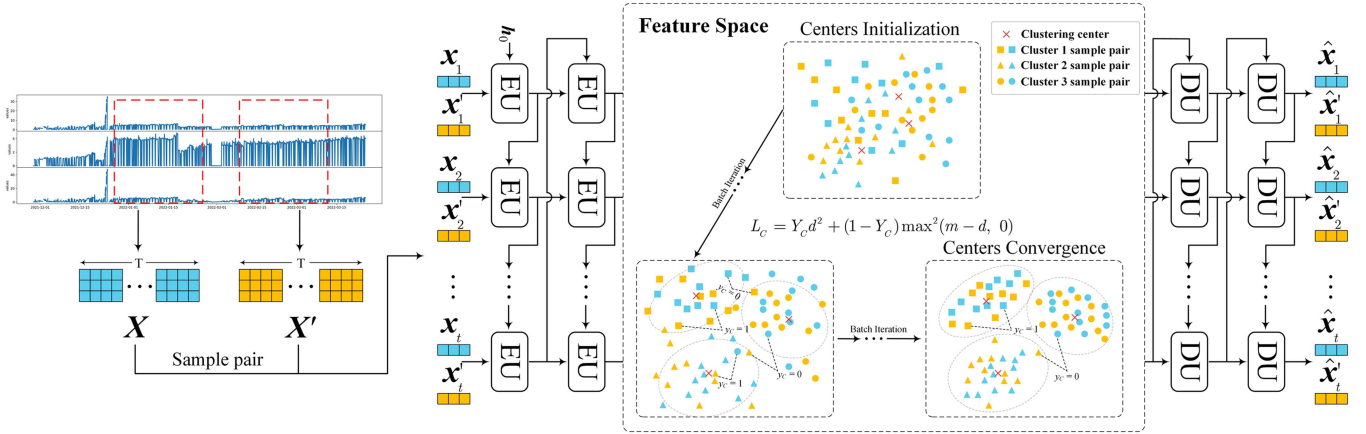


Fig. 2. Architecture of CBCL-AE.

Algorithm 1: Iteration of CBCL.

Input: Hidden states set
 $H = \{h_{t1}^{(2)}, h_{t1}'^{(2)}, \dots, h_{tn}^{(2)}, h_{tn}'^{(2)}, \dots, h_{tN}^{(2)}, h_{tN}'^{(2)}\}$ in i -th batch, distance function $D(\cdot)$
Output: Label set $Y = \{y_1, y_2, \dots, y_n, \dots, y_N\}$

```

1 if  $i = 1$  then
2    $C_i \leftarrow$  randomly select  $k$  hidden states from  $H$ 
3 else
4    $C_i \leftarrow$  update cluster centers of  $H$  with k-means clustering
5    $C_i \leftarrow \frac{C_i + C_{i-1}}{2}$ 
6 end
7 for  $n \leftarrow [1, N]$  do
8   for  $k \leftarrow [1, K]$  do
9      $d_k \leftarrow D(h_{tn}^{(2)}, c_k), d'_k \leftarrow D(h_{tn}'^{(2)}, c_k)$ 
10  end
11   $D \leftarrow \{d_1, d_2, \dots, d_K\}, D' \leftarrow \{d'_1, d'_2, \dots, d'_K\}$ 
12   $k^* \leftarrow \arg \min(D), k'^* \leftarrow \arg \min(D')$ 
13   $y_n \leftarrow$  calculate label through Eq. (5)
14 end

```

Assuming there are K working conditions in the data, during the i th batch iteration, we denote the converged cluster centers of this batch as $C_i = \{c_1, c_2, \dots, c_K\}$. Let $H = \{h_{t1}^{(2)}, h_{t1}'^{(2)}, \dots, h_{tn}^{(2)}, h_{tn}'^{(2)}, \dots, h_{tN}^{(2)}, h_{tN}'^{(2)}\}$ represent the set of hidden states obtained after encoding in a batch. To make the distribution of a batch closer to the overall distribution, we try to make N as large as possible. This allows for more stable cluster centers during the iteration. The distance between $h_{tn}^{(2)}$ and the k th cluster center can be defined as $d_k = D(h_{tn}^{(2)}, c_k)$. Similarly, we can calculate the distance d'_k between $h_{tn}'^{(2)}$ and c_k . As a result, the distance sets can be represented as $D = \{d_1, d_2, \dots, d_K\}$ and $D' = \{d'_1, d'_2, \dots, d'_K\}$. Clusters that $h_{tn}^{(2)}$ and $h_{tn}'^{(2)}$ belong to can be determined as $k^* = \arg \min(D)$ and $k'^* = \arg \min(D')$, respectively. Finally, y in (4) can be defined as

$$y = \begin{cases} 1, & \text{if } k^* = k'^* \\ 0, & \text{else.} \end{cases} \quad (5)$$

We present the pseudocode for the iteration of CBCL during the training process in Algorithm 1.

C. Loss Function

The loss function of CBCL-AE can be defined as

$$L(\theta) = (1 - \alpha)L_{\text{mse}}(\theta) + \alpha(L_C(\theta) + L'(\theta)) \quad (6)$$

where α serves as a tunable parameter that balances the tradeoff between the two terms. The first term $L_{\text{mse}}(\theta)$ is an average reconstruction loss of a pair of inputs with the form

$$L_{\text{mse}}(\theta) = \frac{1}{N} \sum_{n=1}^N \left(\frac{(X - f_{\theta}(X))^2 + (X' - f_{\theta}(X'))^2}{2} \right). \quad (7)$$

The primary purpose of $L_{\text{mse}}(\theta)$ is to constrain the difference between the input and the model's reconstruction.

The second term contains two parts. The loss of contrastive learning $L_C(\theta)$ tends to improve the ability of feature extraction by improving the proximity of features from identical working conditions and enhancing the differentiation of features from different working conditions. The constraint part $L'(\theta)$ is used to constrain the distance between cluster centers, ensuring that there is sufficient differentiation in the feature space among different working conditions. In this study, we define $L'(\theta)$ as

$$L'(\theta) = \max \left(\gamma - \frac{1}{K(K-1)} \sum_{k_1=1}^K \sum_{k_2=1, k_1 \neq k_2}^K D(c_{k_1}, c_{k_2}), 0 \right)^2 \quad (8)$$

where γ is a tunable parameter that limits a minimum average distance and can be determined with a random search algorithm, $\frac{1}{K(K-1)} \sum_{k_1=1}^K \sum_{k_2=1, k_1 \neq k_2}^K D(c_{k_1}, c_{k_2})$ represents the average distance between all cluster centers, excluding the distance from a cluster center to itself.

D. Convergence of Clustering Process

The convergence of the clustering process in the feature space under different working conditions is very important for the model's performance. In general, the clustering process can always reach convergence thanks to the nature of clustering

algorithms. However, according to (4) and Algorithm 1, it can be inferred that the cluster center iteration depends on the features obtained from the CBCL-AE model. For this reason, once the features obtained from the model are unstable, it becomes difficult for the clustering process to converge. Therefore, if the convergence of the model during the training process does exist, we infer that the CBCL-AE model can extract stable features. Consequently, the iteration of cluster centers based on these features will also converge.

For the theoretical proof of convergence, we need two assumptions. For two sets of parameters from the CBCL-AE model, $\forall \theta_1, \theta_2$, obtained from any two iterations during the training process, they satisfy the following condition:

$$\mathbf{A1} : \quad \|\theta_1 - \theta_2\|_2 \leq d \quad (9)$$

where d represents the distance boundary between θ_1 and θ_2 . Since θ_1 and θ_2 are obtained through the stochastic gradient descent (SGD) algorithm, **A1** holds for any differentiable function. Another assumption is the boundedness of gradients for any iteration, $\forall i$, during the training process. For the i th iteration, the gradient g_i satisfies

$$\mathbf{A2} : \quad \|g_i\|_2 \leq d_g. \quad (10)$$

In fact, for any trainable deep learning model, once a local optimum exists during the training process, this local region can satisfy the above-mentioned first assumptions. In addition, we employ the gradient clipping, Xavier initialization, and batch normalization methods during the training to ensure that the second assumption regarding the boundedness of gradients is satisfied. Intuitively, this local optimum region can also guarantee the convexity of the loss function. We measure the convergence by introducing the regret conception in Zinkevich's [28] work with the form

$$C(N) = \sum_{i=1}^N (L_i(\theta_i) - L(\theta^*)) \quad (11)$$

where θ_i denotes the parameters of the CBCL-AE model from the i th iteration during the training process, θ^* represents the parameters that optimizes the loss function $L(\cdot)$. When $N \rightarrow \infty$, if there exists $C(N)/N \rightarrow 0$, it implies that $L_i(\theta_i)$ is approaching $L(\theta^*)$, indicating that the loss function $L(\cdot)$ converges during the training process. Therefore, $\lim_{N \rightarrow \infty} \frac{C(N)}{N} = 0$ serves as the criterion for determining the convergence of the loss function $L(\cdot)$.

For $L(\cdot)$, in the local optimum region that can guarantee the convexity, there exists

$$L_i(\theta_i) - L_i(\theta^*) \leq \langle g_i, \theta_i - \theta^* \rangle. \quad (12)$$

According to (11) and (12), we derive the following inequality:

$$C(N) \leq \sum_{i=1}^N \langle g_i, \theta_i - \theta^* \rangle. \quad (13)$$

Therefore, our objective shifts to providing the boundedness of $\sum_{i=1}^N \langle g_i, \theta_i - \theta^* \rangle$. Based on the stochastic SGD used in the

training process, we have

$$\begin{aligned} \theta_{i+1} &= \theta_i - \eta_i g_i \\ \Rightarrow \theta_{i+1} - \theta^* &= \theta_i - \theta^* - \eta_i g_i \\ \Rightarrow \|\theta_{i+1} - \theta^*\|_2^2 &= \|\theta_i - \theta^* - \eta_i g_i\|_2^2 \\ \Rightarrow \|\theta_{i+1} - \theta^*\|_2^2 &= \|\theta_i - \theta^*\|_2^2 - 2\eta_i \langle g_i, \theta_i - \theta^* \rangle + \eta_i^2 \|g_i\|_2^2 \\ \Rightarrow \langle g_i, \theta_i - \theta^* \rangle &= \frac{1}{2\eta_i} (\|\theta_i - \theta^*\|_2^2 - \|\theta_{i+1} - \theta^*\|_2^2) + \frac{\eta_i}{2} \|g_i\|_2^2. \end{aligned} \quad (14)$$

Therefore, (13) can be further expressed as

$$C(N) \leq \sum_{i=1}^N \frac{1}{2\eta_i} (\|\theta_i - \theta^*\|_2^2 - \|\theta_{i+1} - \theta^*\|_2^2) + \sum_{i=1}^N \frac{\eta_i}{2} \|g_i\|_2^2. \quad (15)$$

According to the assumption (A2) on the boundedness of gradients, the second term in (15) can be transformed into the following inequality:

$$\sum_{i=1}^N \frac{\eta_i}{2} \|g_i\|_2^2 \leq \sum_{i=1}^N \frac{\eta_i d_g^2}{2} = \frac{d_g^2}{2} \sum_{i=1}^N \eta_i. \quad (16)$$

For the first term in (15), we expand it and obtain

$$\begin{aligned} &\sum_{i=1}^N \frac{1}{2\eta_i} (\|\theta_i - \theta^*\|_2^2 - \|\theta_{i+1} - \theta^*\|_2^2) \\ &= \frac{1}{2\eta_1} \|\theta_1 - \theta^*\|_2^2 - \frac{1}{2\eta_1} \|\theta_2 - \theta^*\|_2^2 \\ &\quad + \frac{1}{2\eta_1} \|\theta_2 - \theta^*\|_2^2 - \frac{1}{2\eta_1} \|\theta_3 - \theta^*\|_2^2 + \dots \\ &\quad + \frac{1}{2\eta_N} \|\theta_N - \theta^*\|_2^2 - \frac{1}{2\eta_N} \|\theta_{N+1} - \theta^*\|_2^2 \\ &= \frac{1}{2\eta_1} \|\theta_1 - \theta^*\|_2^2 + \sum_{i=2}^N \left(\frac{1}{2\eta_i} - \frac{1}{2\eta_{i-1}} \right) \|\theta_i - \theta^*\|_2^2 \\ &\quad - \frac{1}{2\eta_N} \|\theta_{N+1} - \theta^*\|_2^2. \end{aligned} \quad (17)$$

Based on the assumption (A1) on the boundedness of parameters and the monotonically nonincreasing nature of the learning rate η during the training process, we obtain the following three inequalities:

$$\frac{1}{2\eta_1} \|\theta_1 - \theta^*\|_2^2 \leq \frac{1}{2\eta_1} d^2 \quad (18)$$

$$\sum_{i=2}^N \left(\frac{1}{2\eta_i} - \frac{1}{2\eta_{i-1}} \right) \|\theta_i - \theta^*\|_2^2 \leq d^2 \sum_{i=2}^N \left(\frac{1}{2\eta_i} - \frac{1}{2\eta_{i-1}} \right) \quad (19)$$

$$- \frac{1}{2\eta_N} \|\theta_{N+1} - \theta^*\|_2^2 \leq 0. \quad (20)$$

Algorithm 2: Training procedure of CBCL-AE.

Input: Training sample set $D_{\mathbf{X}} = \{(\mathbf{X}_1, \mathbf{X}'_1), (\mathbf{X}_2, \mathbf{X}'_2), \dots, (\mathbf{X}_S, \mathbf{X}'_S)\} \in \mathbb{R}^{S \times m \times T}$, the iterations number I , learning rate η , the SGD optimization function $g(\nabla \theta_i, \eta)$, loss function $L(\theta)$, early stopping parameter $v = 0$, $u = 0$

Output: trained model's parameters θ^*

```

1 Initialize the parameters of the CBCL-AE as  $\theta_0$ 
2 for  $i \leftarrow [1, I]$  do
3   feed  $D_{\mathbf{X}}$  into the CBCL-AE in batch
4    $\hat{\mathbf{X}} \leftarrow$  compute the reconstruction
5    $\mathbf{Y} \leftarrow$  compute the label set with Algorithm 1
6    $L(\theta) \leftarrow$  compute the loss
7    $\nabla L(\theta) \leftarrow$  compute the gradient  $\frac{\partial L(\theta)}{\partial \theta}$ 
8    $\theta_i \leftarrow$  update parameters with  $g(\nabla L(\theta_{i-1}), \eta)$ 
9   if  $L(\theta_i) - L(\theta_{i-1}) \leq \varepsilon$  then
10    |  $u+ = 1$ 
11   else
12    |  $u = 0$ 
13   end
14   if  $u \geq v$  then
15    | return trained parameters  $\theta^*$  and the trained CBCL-AE
16   end
17 end
18 return trained parameters  $\theta^*$  and the trained CBCL-AE

```

Taken (17)–(20) together, we have

$$\sum_{i=1}^N \frac{1}{2\eta_i} \left(\|\theta_i - \theta^*\|_2^2 - \|\theta_{i+1} - \theta^*\|_2^2 \right) \leq \frac{1}{2\eta_1} d^2 + d^2 \sum_{i=2}^N \left(\frac{1}{2\eta_i} - \frac{1}{2\eta_{i-1}} \right) + 0 = \frac{d^2}{2\eta_N}. \quad (21)$$

Finally, we derive

$$C(N) \leq \frac{d^2}{2\eta_N} + \frac{d_g^2}{2} \sum_{i=1}^N \eta_i. \quad (22)$$

Since both d and η_i are constant, and η_i is monotonically nonincreasing, we can conclude that regret $C(N)$ is bound. Therefore, as $\lim_{N \rightarrow \infty} \frac{C(N)}{N} = 0$, $L(\cdot)$ converges during the training process. In this case, the convergence of $L(\cdot)$ implies the features obtained from CBCL-AE tend to be stable. Therefore, we can conclude that the feature clustering process also converges. This convergence can also be empirically confirmed by visualizing the trace of cluster center iteration in Section IV-C.

E. Optimization Procedure

Once the final loss function is determined, the SGD algorithm is used to train the model by optimizing

$$\theta^* = \arg \min_{\theta} ((1 - \alpha)L_{\text{mse}}(\theta) + \alpha(L_C(\theta) + L'(\theta))). \quad (23)$$

We present the detailed optimization procedure in Algorithm 2.

IV. EXPERIMENTAL DESIGN AND RESULTS

A. Experimental Design

1) **Baseline Models:** To validate the effectiveness of our model, we compare the proposed CBCL-AE model with five other SOTAs on three public datasets and two private

datasets. The five SOTAs are Anomaly-Transformer [23], OmniAnomaly [20], long short-term memory variational autoencoder (LSTM-VAE) [21], encoder-decoder anomaly detection (EncDec-AD) [19], and unsupervised anomaly detection model (USAD) [29], respectively.

Among the above-mentioned SOTAs, Xu et al. [23] utilized the encoder component from the transformer architecture to build the Anomaly Transformer model, which utilizes a specific association discrepancy constraint to enhance the model's ability to handle MTS. Su et al. [20] extended the GRU-VAE model by incorporating temporal dependencies into the latent variables, resulting in the creation of OmniAnomaly. Park et al. [21] expanded upon the LSTM autoencoder with variational inference techniques to improve the robustness. EncDec-AD [19] is an LSTM-based autoencoder, while USAD [29] integrates the generative adversarial strategy into a traditional autoencoder.

2) **Threshold Selection:** After obtaining the reconstruction results based on the CBCL-AE model, it is necessary to calculate the residuals, $e = \mathbf{X} - \hat{\mathbf{X}}$, using the input data and the reconstruction results. The calculated residuals can thus be used to identify anomalies with a given threshold.

To ensure fairness, we employ the same threshold selection strategy for all models. First of all, we generate a sufficient number of thresholds within the range of residuals. Then, we select a threshold that yields the optimal result for anomaly detection. Considering the capability of CBCL-AE to adaptively recognize working conditions, we select different thresholds for different working conditions. In other words, the threshold selection process will be repeated for each working condition.

In addition, we adopt the same anomaly detection rule for all models. The rule is relatively simple: an anomaly is identified if the residual value is greater than the threshold.

3) **Evaluation Metrics:** Due to the significant imbalance between normal data and anomalies, accuracy is not suitable for evaluating the model performance. Therefore, we employ the evaluation metrics, such as precision, recall, and F1-score, which are defined as follows:

$$\begin{aligned} \text{precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\ \text{recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{F1-score} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \quad (24)$$

where TP represents true positives, FP represents false positives, and FN represents false negatives. Since the F1-score is the harmonic mean of precision and recall, we concentrate on and use it as the main evaluation metric.

B. Experimental Results

1) **Data:** We consider three public datasets: Server machine datasets (SMD) by Su et al. [20], Soil Moisture Active Passive (SMAP) satellite, and Curiosity Rover on Mars (MSL) by citeRN22, as well as our two private datasets: Dataset of multisensor on pump (MSPD) and dataset of multisensor on fan

TABLE I
DETAILED INFORMATION ON DATASETS

Name	Entities	Dimensions	Training size	Testing size	Anomaly ratio (%)
SMD	28	38	708 405	708 420	4.16
MSL	27	55	58 317	73 729	10.72
SMAP	55	25	135 183	427 617	13.13
MSPD	75	3	150 201	105 945	1.18
MSFD	17	3	38 720	29 345	1.43

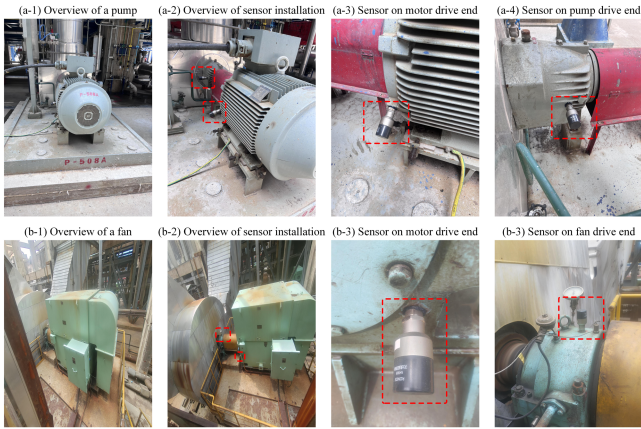


Fig. 3. Installation of sensors. (a) Sensors on a pump. (b) Sensors on an induced draft fan.

(MSFD). These datasets detailed in Table I are used to demonstrate the superiority of CBCL-AE in unsupervised anomaly detection. The three public datasets do not exhibit the nature of variable working conditions and are considered as general MTS in this study. They are used to demonstrate that CBCL-AE can achieve satisfactory anomaly detection performance on general MTS.

The MSPD and MSFD are both 3-D MTS data. The former is collected from multiple sensors on a pump, and the latter is from an induced draft fan. The three dimensions represent the root-mean-square of velocity, the energy at the fundamental frequency, and the energy at twice the fundamental frequency, respectively. These three metrics are primarily used to detect rotation anomalies in bearings, such as imbalance and misalignment. Here, the fundamental frequency indicates the rotational frequency. The installation of multiple sensors on a machine (e.g., pump, induced draft fan) is illustrated in Fig. 3. The MSPD consists of a training set and a test set. The training set comprises normal data from 40 pumps, while the test set contains 1251 anomalies collected from 35 pumps. Similarly, the training set of MSFD comprises normal data from ten induced draft fans, while the test set contains 420 anomalies collected from seven induced draft fans. In these two datasets, the time interval between two consecutive data points is two hours in both the training and test sets. The working condition is measured by the three metrics mentioned previously. We visualize the data collected from a specific pump and an induced draft fan in Fig. 4, where three different working conditions are marked with different colors. Obviously, the MTS evolves and presents

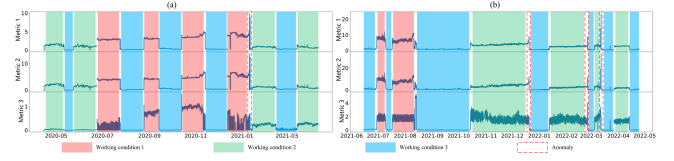


Fig. 4. Data visualization of private datasets. (a) MSPD. (b) MSFD.

TABLE II
COMPARISON RESULTS ON THREE PUBLIC DATASETS

Model	SMD			MSL			SMAP		
	P	R	F1	P	R	F1	P	R	F1
AT	0.8989	0.9701	0.9331	0.8043	0.9889	0.8871	0.9103	0.9597	0.9343
OmniAnomaly	0.8332	0.8662	0.8494	0.9130	0.8255	0.8670	0.8887	0.8741	0.8813
LSTM-VAE	0.7688	0.8902	0.8251	0.9288	0.6699	0.7784	0.8559	0.7890	0.8211
EncDec-AD	0.9180	0.7901	0.8493	0.7400	0.7098	0.7246	0.8233	0.8041	0.8136
USAD	0.7031	0.5959	0.6451	0.5587	0.6156	0.5858	0.4912	0.5993	0.5399
GRU-AE	0.8847	0.8062	0.8436	0.7641	0.7233	0.7431	0.8307	0.8246	0.8276
CBCL-AE	0.8988	0.7963	0.8445	0.7631	0.6988	0.7295	0.8565	0.7966	0.8255

Note: (1) P, R, and F1 denote the metrics of precision, recall, and F1-score, respectively. (2) Boldface represents the best result in each column while the underline represents the suboptimal result. (3) AT represents the Anomaly Transformer model.

several abrupt changes in mean and variance values. This implies that working conditions are not fixed but may change from one to another, also called variable working conditions.

Since we have prior knowledge of the working condition labels for machines on these two private datasets, this study has three working conditions. Therefore, we fix this hyperparameter, the number of clustering centers K , to three in the CBCL module for subsequent experiments. However, in general scenarios where the number of working conditions is unknown, we should treat K as a hyperparameter and select it through sensitivity analysis.

The main difference between the three public datasets and the two private datasets (e.g., the MSPD and MSFD) is that the public datasets do not have the nature of variable conditions, namely their distribution, including mean and variance, remains relatively stable over time.

2) Comparison Results: To demonstrate that CBCL-AE has satisfactory anomaly detection performance for general MTS, we conduct experiments on three public datasets: SMD, MSL, and SMAP. We compare it with the five SOTAs and report the comparison results in Table II, with the average value across ten repeated experiments.

Table II shows that although the CBCL-AE model does not perform best among the five SOTAs on the three public datasets, it still achieves the average level. As for the F1-score, we can observe that the CBCL-AE model exhibits the fourth-best performance on both SMD (F1-score=0.8445) and MSL (F1-score=0.8255) while achieving only the fifth-best on SMAP (F1-score=0.7295). Specifically, although the CBCL-AE model ranks fourth in the F1-score on SMD and MSL, its performance does not differ significantly from the models that are suboptimal or third best. Therefore, we believe that the CBCL-AE model achieves acceptable anomaly detection performance on the three public datasets. We also find that the model GRU-AE, without the CBCL module, maintains acceptable anomaly detection results on those public datasets with the F1-scores of 0.8436, 0.7431, and 0.8276, respectively.

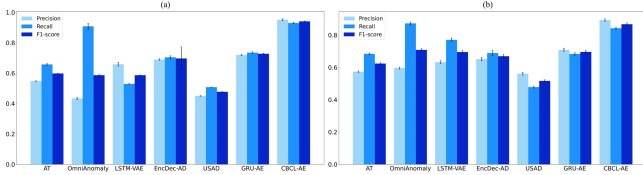


Fig. 5. Comparison results on the MSPD.

TABLE III
PAIRED T-TEST RESULTS ON TWO PRIVATE DATASETS

H_1	t-statistic on the MSPD	t-statistic on the MSFD
$\mu_1 > \mu_2$	291.7968 ***	82.6425 ***
$\mu_1 > \mu_3$	204.9337 ***	47.7802 ***
$\mu_1 > \mu_4$	288.1122 ***	74.3283 ***
$\mu_1 > \mu_5$	11.7392 ***	56.6770 ***
$\mu_1 > \mu_6$	342.6412 ***	115.8213 ***
$\mu_1 > \mu_7$	112.6211 ***	47.8567 ***

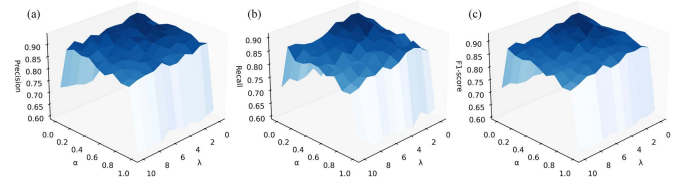
Note: (1) The subscript numbers 1–7 represent different models. 1: CBCL-AE; 2: Anomaly Transformer; 3: OmniAnomaly; 4: LSTM-VAE; 5: EncDec-AD; 6: USAD; 7: GRU-AE. (2) *, **, and *** denote statistically significant at the 10%, 5%, and 1% level, respectively.

Since the three public datasets do not exhibit the typical nature of variable working conditions, we further conduct experiments on our private datasets: the MSPD and MSFD, to demonstrate the advantages of CBCL-AE in handling data under variable working conditions. We compare the CBCL-AE model with the five SOTAs on these two private datasets and report the average values with standard deviations across ten repeated experiments in Fig. 5.

It is evident from Fig. 5 that the CBCL-AE model exhibits remarkable superiority on the two private datasets over the other SOTAs in terms of three key metrics: precision (0.951 and 0.8931), recall (0.9302 and 0.8425), and F1-score (0.9407 and 0.8671). Moreover, the CBCL-AE model also exhibits stable performance, as evidenced by its low standard deviations in these two datasets. In addition, the GRU-AE model without the CBCL module is far inferior to the CBCL-AE model. These comparison and ablation study results strongly demonstrate the capability of CBCL-AE in handling data under variable working conditions, primarily attributed to the utilization of the CBCL module.

We further use the paired t-test to test whether the superiority of the CBCL-AE model to the others is statistically significant. The alternative hypothesis (H_1) is that the CBCL-AE model is superior to another in terms of greater F1-score, i.e., $\mu_1 > \mu_j$ for $j = 2, 3, \dots, 7$. In this case, μ_i denotes the average F1-score of model i ($i = 1, 2, \dots, 7$) across ten repetitions. We present the test results in Table III and find that all tests are statistically significant at the 1% level. Thus, we conclude that the CBCL-AE model significantly outperforms the other models in unsupervised anomaly detection on two private datasets.

In summary, although the CBCL-AE model is not optimal on the three public datasets, it delivers a satisfactory performance, indicating its acceptable anomaly detection capability for general MTS. More important experiments on the MSPD and MSFD illustrate the superiority of CBCL-AE and demonstrate its unique capability to detect anomalies in MTS under variable working conditions.

Fig. 6. Sensitivity analysis of α and λ . (a) Precision. (b) Recall. (c) F1-score.

C. Model Discussion

Since the MSPD provides a more extensive amount of data and shows more typical natures of variable working conditions than the MSFD, we further discuss the model performance of the MSPD by conducting sensitivity analysis, model complexity, reconstruction evaluation, verification of the convergence of the clustering process, and evaluation of the model's ability to recognize working conditions.

1) *Sensitivity Analysis of Hyperparameters*: In the loss function of CBCL-AE, we introduce the tunable hyperparameter that balances the trade-off between two different losses, see (6). In addition, hyperparameter λ in the contrastive learning loss limits the minimum feature distance between the observations from different working conditions, see (4). To demonstrate the impact of these two hyperparameters on the CBCL-AE model, we conduct a sensitivity analysis and visualize the results in Fig. 6. We find that the CBCL-AE model achieves optimal performance on all three metrics (precision, recall, F1-score) when the hyperparameters are selected around $\alpha = 0.2$ and $\lambda = 2$. In particular, the model's performance significantly deteriorates when $\alpha = 0$ or $\alpha = 1$. This is because the model does not integrate the CBCL module when $\alpha = 0$. In this case, the model is unable to recognize working conditions adaptively. In addition, when $\alpha = 1$, the model essentially removes the MSE loss function, which affects its crucial reconstruction capability and weakens its anomaly detection ability. According to the sensitivity analysis, we adopt the hyperparameter combination $\alpha = 0.2$ and $\lambda = 2$ in our experiment.

2) *Model Complexity*: Since the CBCL-AE model integrates the CBCL module; the increase in computational burden is unavoidable. However, this module is just introduced during the training to enlarge the difference between features across different working conditions. Therefore, the additional computational burden of CBCL-AE mainly occurs during the training phase. Once the model completes training, no extra computation is required during the online anomaly detection. To show the model complexity of CBCL-AE, we compare the memory usage, training time for each batch, and testing time for each batch on the MSPD against all comparison models. According to Table IV, although the CBCL-AE model takes longer to train, it ranks second in testing time among all models. This suggests that the CBCL-AE model is well-suited for online real-time anomaly detection.

3) *Evaluation of Reconstruction*: In the reconstruction-based anomaly detection model, the quality of normal data reconstruction indicates their capability for learning the normal behavior of the data. In other words, when a model effectively reconstructs

TABLE IV
COMPARISON OF MODEL COMPLEXITY

Model	Memory usage (KB)	Training time/batch (s)	Testing time/batch (s)
AT	28604	0.1075	0.0284
OmniAnomaly	19660	0.6791	0.3520
LSTM-VAE	150	0.1137	0.0389
EncDec-AD	125	0.0278	0.0090
USAD	9498	0.5307	0.0020
GRU-AE	120	0.0271	0.0080
CBCL-AE	120	4.9867	0.0070

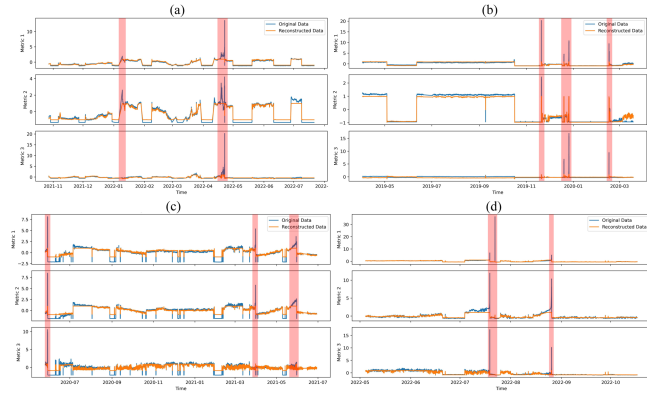


Fig. 7. Visualizations of data reconstruction. (a) Pump 1. (b) Pump 2. (c) Pump 3. (d) Pump 4.

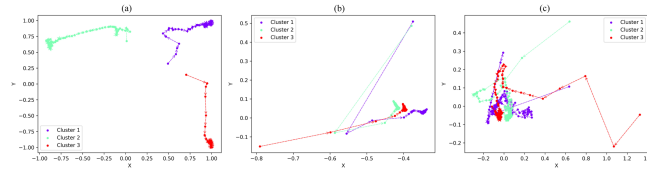


Fig. 8. Iterations of cluster centers. (a) Iterations of cluster centers of CBCL-AE. (b) Iterations of cluster centers without $L'(\theta)$. (c) Iterations of cluster centers without CBCL.

normal data while having poor reconstruction quality for anomalies, it enables the differentiation between the reconstructions and actual anomalies. This helps to improve anomaly detection by leveraging such differentiation. To evaluate the reconstruction capability, we visualize the reconstruction results from the CBCL-AE model in Fig. 7, where the highlighted red areas represent anomalies. The reconstruction results indicate that the CBCL-AE model has effectively learned the normal behavior of the data. This is evident in Fig. 7, where normal data can be effectively reconstructed, while anomalies are difficult to reconstruct. This results in larger reconstruction residuals for abnormal instances, enabling accurate anomaly detection.

4) Visualization of the Clustering Process: Since the CBCL module is introduced into the CBCL-AE model, it is necessary to verify the convergence of the clustering process during the training process. In addition to the theoretical proof in Section III-D, we further confirm the convergence of the clustering process by visualizing the trace of the cluster center iteration in Fig. 8, where the X and Y coordinates represent two dimensions of the

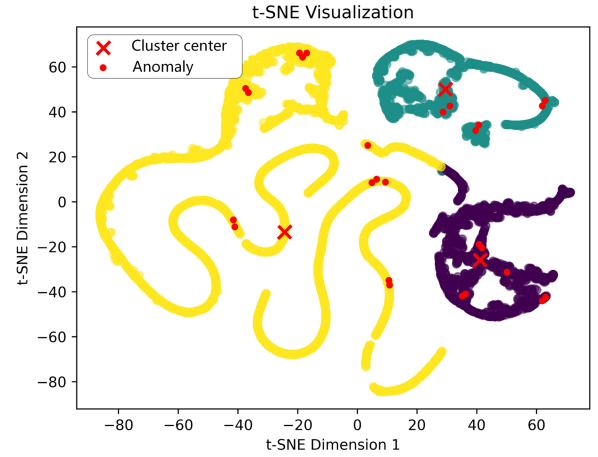


Fig. 9. t-SNE visualization of the feature space.

feature space. Specifically, Fig. 8(a) illustrates the trace of cluster center iteration of CBCL-AE. We observe that the three clusters with three different colors, representing three working conditions, gradually converge. Moreover, the final cluster centers maintain sufficient distance from each other, indicating that the clusters can be effectively distinguished. In Fig. 8(b), although the iteration of final cluster centers also maintains convergence, the absence of distance constraint, $L'(\theta)$, leads to a lack of differentiation among the converged centers. This is reflected in the fact that the converged cluster centers are close to each other. In addition, once removing the CBCL constraints in (6), Fig. 8(c) shows that the clustering process is no longer converged. This means that the model loses the capability to adaptively recognize different working conditions.

5) Evaluation of Working Conditions Recognition: To evaluate the effect in adaptively recognizing variable working conditions, we present the t-SNE visualization in the feature space of CBCL-AE in Fig. 9. The three colors represent three different working condition features in the feature space. The cluster centers of the three different working conditions are marked with crosses. The visualization results show that the three working conditions are well separated in the feature space and features under the same working condition tend to cluster together. This indicates that our proposed CBCL-AE model is indeed capable of recognizing variable working conditions adaptively. In addition, we also mark the features of some anomalies in the feature space with red dots. It is obvious that the features of these anomalies are mostly distributed in the feature space of normal data. This means that the CBCL-AE model can capture the characteristics of normal behavior effectively. This ability is good for the model to reconstruct abnormal inputs as normal ones, thereby amplifying the residual results of anomalies and making them easier to detect.

6) Strengths and Weaknesses: The CBCL-AE model has at least three strengths. First, it leverages the CBCL module to adaptively cluster different working conditions and enlarge the difference between features across these working conditions. Second, it not only achieves satisfactory performance in general MTS but, more importantly, performs excellently on MTS data

with the typical nature of working conditions. Third, it can be deployed in online anomaly detection without excessive computational resources. In addition, the CBCL-AE model also has two weaknesses. Although it achieves satisfactory performance on general MTS without the nature of variable working conditions, such performance still has room for improvement compared to the best SOTA. In addition, its training process requires more computational resources compared to other models.

V. CONCLUSION

To achieve unsupervised anomaly detection for machines under variable working conditions, we propose a novel CBCL-AE. This model integrates CBCL to cluster data in the feature space and can effectively improve the proximity of features from identical working conditions while enhancing the differentiation of features from different working conditions. By leveraging the CBCL, our proposed model can achieve adaptive working condition recognition. Furthermore, a specific loss function is designed for CBCL-AE to ensure its effectiveness. We theoretically prove the convergence property of the clustering process of CBCL during the training process, which directly determines the effectiveness of CBCL-AE.

Due to the absence of variable working conditions in the three public datasets (SMD, SMAP, and MSL), our CBCL-AE model achieves acceptable anomaly detection performance on general MTS compared to the five SOTAs. Moreover, the experimental results on the two private datasets, namely the MSPD and MSFD, show that the CBCL-AE model performs best and even achieves an impressive F1-score of up to 0.9407 on MSPD and 0.8671 on MSFD, respectively. The ablation study strongly validates the role of CBCL in improving the performance of the CBCL-AE model. Visualizing the trace of cluster center iteration during the training process further confirms the convergence of the clustering process and demonstrates the effectiveness of CBCL as well as the role of the loss $L'(\theta)$. In addition, the t-SNE visualization demonstrates that the CBCL-AE model has an excellent capability of adaptive working conditions recognition and learning normal behavior.

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Qifa Xu received the B.E. degree in mathematics education from the Department of Mathematics, Fuyang Normal University, Fuyang, China, in 1997, the M.S. degree in quantitative economics from the Dongbei University of Finance and Economics, Dalian, China, in 2000, and the Ph.D. degree in technical economics and management from the School of Management, Tianjin University, China, in 2006.

He is currently a Professor with the School of Management, Hefei University of Technology, Hefei, China. His research interests include financial big data analysis, statistical learning, and smart manufacturing.



Qiliang Cheng received the B.S. and M.S. degrees in business management from Hefei University of Technology, Hefei, China, in 2018 and 2022, respectively.

He is currently with the Intelligent Algorithm Department, Anhui RONS Science and Technology Incorporated Company, a company that provides monitoring and maintenance solutions and services for various industries. His research focuses on bearing fault diagnosis.



Tianming Xie received the B.S. degree in business management from the Hefei University of Technology, Hefei, China, in 2019. He is currently working toward the Ph.D. degree in management science and engineering with the School of Management, Hefei University of Technology.

His research interests include anomaly detection and fault diagnosis, and their applications in industrial equipment, and renewable energy.



Xiangxiang Wang is currently working toward the Ph.D. degree in management science and engineering with the School of Management, Hefei University of Technology, Hefei, China.

She is the Deputy Minister of Anhui RONS Science and Technology Incorporated Company. She has extensive experience in diagnosing components, such as bearings and gearboxes. Her research interests include data-driven fault diagnosis and prognosis, reliability engineering, and edge computing.



Cuixia Jiang received the B.E. degree in mathematics education from the Department of Mathematics, Fuyang Normal University, Fuyang, China, in 1997, and the Ph.D. degree in technical economics and management from the School of Management, Tianjin University, Tianjin, China, in 2008.

She is currently an Associated Professor with the School of Management, Hefei University of Technology, Hefei, China. Her research interests include financial big data analysis, financial

time series analysis, and statistical learning.